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| <p><b>Text Book</b></p> <p>Carl Hamacher, Zvonko Vranesic, Safwat Zaky, “Computer Organization”, Fifth Edition, McGraw-Hill Higher Education, 2016 reprint.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                           |         |     |   |   |
| <p><b>References</b></p> <p>William Stallings, “Computer Organization &amp; Architecture – Designing for Performance”, 11th Edition, Pearson Education Inc., 2019</p> <p>David A. Patterson &amp; John L. Hennessy, “Computer Organization and Design MIPS Edition- The Hardware/Software Interface”, 6th Edition, Morgan Kaufmann, Elsevier Publications, November 2020.</p> <p><b>Web References:</b></p> <p>NPTEL Course on “Computer architecture and organization” IIT Kharagpur By Prof. Indranil Sengupta, Prof. Kamalika Datta. <a href="https://nptel.ac.in/courses/106105163">https://nptel.ac.in/courses/106105163</a></p> <p>NPTEL Course on “Computer Organization”, IIT Madras By Prof. S. Raman. <a href="https://nptel.ac.in/courses/106106092">https://nptel.ac.in/courses/106106092</a></p> <p><a href="https://puniversity.informaticsglobal.com:2229/login.aspx">https://puniversity.informaticsglobal.com:2229/login.aspx</a></p> |                                           |         |     |   |   |
| <p>Topics relevant to “SKILL DEVELOPMENT”: Generation of Computers, CISC and RISC processors, Bus Arbitration, Collaboration and Data collection for Term assignments and Case Studies for Skill Development through Participative Learning techniques. This is attained through assessment component mentioned in course handout.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                           |         |     |   |   |
| Course Code:<br>CSE203                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Course Title: Discrete Mathematics        | L-T-P-C | 4-0 | 0 | 4 |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Type of Course: Program Core& Theory Only |         |     |   |   |
| Version No.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 2.0                                       |         |     |   |   |
| Course Pre-requisites                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | NIL                                       |         |     |   |   |
| Anti-requisites                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | NIL                                       |         |     |   |   |

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| Course Description                                                                                                                                                                                                                             | This course highlights the basics of discrete structures and develop ability to solve problems involving mathematical logic, sets, functions, relations, principles of counting, pigeonhole principles, recurrence relations, Principles of Inclusion and Exclusion. forces, and moments with their applications in allied subjects. It is a prerequisite for several Courses involving Compiler Design, Artificial Intelligence. This course is both conceptual and analytical in nature that would help the student to use the concepts of discrete structures to solve and prediction of data analytics. The students should have prior knowledge of basic mathematics pursue the Course. After successful completion of the Course, the students would acquire knowledge to solve problems involving mathematical logic, sets, functions, relations, principles of counting, pigeon hole principles, recurrence relations, Principles of Inclusion and Exclusion with an emphasis on real-world engineering applications and problem solving. |            |                 |             |
| Course Objective                                                                                                                                                                                                                               | The objective of the course is to familiarize the learners with the concepts of Discrete Mathematics and attain SKILL DEVELOPMENT through PROBLEM SOLVING Methodologies techniques.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |            |                 |             |
| Course Out Comes                                                                                                                                                                                                                               | <p>On successful completion of the course the students shall be able to:</p> <p>1] Describe a logic sentence in terms of predicates, quantifiers, and logical connectives.</p> <p>2] Solve problems on Functions and Relations using basic principles of Set Theory.</p> <p>3] Explain the concepts of Boolean Algebra.</p> <p>4] Apply basic counting techniques to combinatorial problem.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |            |                 |             |
| Course Content:                                                                                                                                                                                                                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |            |                 |             |
| Module 1                                                                                                                                                                                                                                       | Foundations of Logics and Proofs                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Assignment | Problem Solving | 10 Sessions |
| <p>Topics:</p> <p>Propositional Logic, Propositional Logic Equivalences, Inference rules, Normal forms, Introduction to Proofs, Resolution by Refutation, Predicates and Quantifiers, Introduction to Proofs.</p> <p>Assignment: Problems.</p> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |            |                 |             |
| Module 2                                                                                                                                                                                                                                       | Basic Structures:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Assignment | Problem Solving | 10 Sessions |

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|                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Sets, Functions, Relations           |            |                 |             |
| <p>Topics:</p> <p>Sets and set-operations, Venn Diagram, Cardinality of Sets, Functions: Types, Invertible Functions, Composition, Sequences and Summations, Relations and their properties &amp; representations, Equivalence Relations, Closure of Relations.</p> <p>Assignment: Problems and applications</p>                                                                                                                                  |                                      |            |                 |             |
| Module 3                                                                                                                                                                                                                                                                                                                                                                                                                                          | Posets, Lattices and Boolean Algebra | Assignment | Problem Solving | 10 Sessions |
| <p>Topics:</p> <p>Partial ordering, Poset, Hasse Diagram, Lattices &amp; Algebraic structures, Basic properties of algebraic systems by lattices, Distributive lattices, complement of an element in a lattice, Boolean lattice &amp; Boolean algebra, Topological Sorting.</p> <p>Assignment: Problems and Applications</p>                                                                                                                      |                                      |            |                 |             |
| Module 4                                                                                                                                                                                                                                                                                                                                                                                                                                          | Principles of Counting Techniques    | Assignment | Problem Solving | 12 Sessions |
| <p>Topics:</p> <p>Number Theory: Integers and Division, GCD, Chinese Remainder Theorem, Solving Congruences, Pigeon Hole Principle, Mathematical Induction, Generalized Permutations and Combinations, Recurrence Relations, Applications of Recurrence Relations, Generating Functions, Principle of Inclusion and Exclusion, Applications of Inclusion and Exclusion.</p> <p>Assignment: Problems and Applications</p>                          |                                      |            |                 |             |
| <p>Targeted Application &amp; Tools that can be used:</p> <p>NIL</p>                                                                                                                                                                                                                                                                                                                                                                              |                                      |            |                 |             |
| <p>Project work/Assignment:</p>                                                                                                                                                                                                                                                                                                                                                                                                                   |                                      |            |                 |             |
| <p>Problems on all the topics and relevance with field of computer science</p>                                                                                                                                                                                                                                                                                                                                                                    |                                      |            |                 |             |
| <p>Text Book</p> <p>T1. Kenneth H. Rosen, "Discrete Mathematics and its Applications", McGraw-Hill, 7th Edition, 2018.</p>                                                                                                                                                                                                                                                                                                                        |                                      |            |                 |             |
| <p>References</p> <p>R1: Susanna EPP, "Discrete Mathematics with Applications", Cengage Learning, 4th Edition, 2010</p> <p>R2. Thomas Koshy, "Discrete Mathematics with Applications", Elsevier, India, 2009.</p> <p>R3: Discrete mathematics for Computer Scientists and Mathematicians, Paperback (Rs. 533), Joel Mott, Abraham Kandel, Theodore Baker; Pearson Education India; 2 edition (2015), ISBN-13: 978-9332550490</p> <p>Weblinks:</p> |                                      |            |                 |             |